# Class 9 Science: Comprehensive Cell Biology Notes

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Class 9 (■■■ ■■■■■) • Science (Biology) • Prepared by: Smt. Shailaja Gowda, Govt High School Faculty

## 1. Discovery of the Cell

Robert Hooke first discovered cells in 1665 using a primitive microscope, observing thin slices of bottle cork. Anton von Leeuwenhoek in 1674 observed living free cells in pond water with an improved microscope. In 1831, Robert Brown discovered the nucleus inside the cell.

## 2. Cell Theory

The cell theory was proposed by Matthias Schleiden (1838) and Theodor Schwann (1839). It states: (a) All living organisms are composed of one or more cells. (b) The cell is the basic structural and functional unit of life. In 1855, Rudolf Virchow added: 'Omnis cellula e cellula' — all cells arise from pre-existing cells.

## 3. Plasma Membrane & Diffusion

The plasma membrane is a flexible, selectively permeable membrane composed of lipids and proteins. Osmosis is the passage of water from a region of high water concentration through a semi-permeable membrane to a region of low water concentration. In hypotonic solutions, cells swell; in hypertonic solutions, cells shrink (plasmolysis).

## 4. Key Cell Organelles

• Mitochondria: Known as the powerhouse of the cell because they synthesize ATP (adenosine triphosphate) molecules through cellular respiration.
• Endoplasmic Reticulum (ER): RER with ribosomes synthesizes proteins; SER manufactures lipids.
• Golgi Apparatus: Packages, modifies, and dispatches materials synthesized in the cell.
• Lysosomes: 'Suicide bags' containing digestive enzymes to breakdown cellular debris.

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